

Limits – 5

Recurrences $x_{n+1} = f(x_n)$: fixed points and stability

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Theoretical minimum.

Def. A point L with $f(L) = L$ is called a fixed point. It is *attracting* if the iterates starting in some neighbourhood converge to L , and *repelling* if every point other than L itself eventually leaves that neighbourhood.

Theorem 1 (stability criterion). Let f be differentiable at L with $f(L) = L$. If $|f'(L)| < 1$, then L is attracting and the convergence is geometric: $|x_n - L| \leq Cq^n$ for every $q \in (|f'(L)|, 1)$. If $|f'(L)| > 1$, then L is repelling. The case $|f'(L)| = 1$ is neutral: there the rate is polynomial and is given by the lemma from the third sheet.

Theorem 2 (monotone iterations). If f is continuous and increasing on an interval I invariant under f , then every sequence $x_{n+1} = f(x_n)$ is monotone and converges to a fixed point. If f is decreasing, the subsequences (x_{2n}) and (x_{2n+1}) are monotone and their limits form a 2-cycle of f , that is, fixed points of $f \circ f$.

Fact 1. The contraction principle: if $|f(x) - f(y)| \leq q|x - y|$ with $q < 1$ on a complete set, then the fixed point is unique, the iterates converge to it from every starting point, and $|x_n - L| \leq \frac{q^n}{1-q}|x_1 - x_0|$.

1. Prove both theorems of the preliminaries and analyse three examples in their language: $x_{n+1} = \cos x_n$, $x_{n+1} = \frac{2}{1+x_n}$ and $x_{n+1} = x_n^2$. In each case list the fixed points, their type and the rate of convergence.
2. Let f be decreasing and suppose $f \circ f$ has exactly one fixed point. Prove that the iterates converge for every initial value; then construct a decreasing f possessing a 2-cycle and describe the behaviour of the iterates in that case.

3. (IMC 2025 shortlist) Let $x_1, y_1, z_1 \in \mathbb{R}$ and $0 < \alpha < 1$, and define

$$x_{n+1} = \alpha x_n + (1-\alpha)y_n, \quad y_{n+1} = \alpha y_n + (1-\alpha)z_n, \quad z_{n+1} = \alpha z_n + (1-\alpha)x_n.$$

Prove that all three sequences converge and find their common limit.

4. (IMC) Define the sequence of continuously differentiable functions $f_1, f_2, \dots : [0, 1) \rightarrow \mathbb{R}$ by the recurrence $f_1 = 1$, $f'_{n+1} = f_n f_{n+1}$ on $(0, 1)$ and $f_{n+1}(0) = 1$. Show that $\lim_{n \rightarrow \infty} f_n(x)$ exists for every $x \in [0, 1)$ and determine the limit function.
5. (ICMC 2022–23) Let $a_1 = 2022$ and $a_{n+1} = a_n + e^{-a_n}$ for $n \geq 1$. Prove that there is a positive real r for which the sequence $\{ra_1\}, \{ra_{10}\}, \{ra_{100}\}, \dots$ converges.